



Interpretable Deep Features for Cell Classification

Report on the Cell Classification Challenge

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1. Introduction

1.1. Context and objective

This report details the methods used to develop a solution for the *IMA205 Challenge 2026*. The goal of this challenge was to build a machine learning model able to efficiently classify white blood cells. Because those cells play a key role in the immune system, being able to sort them as precisely and as quickly as possible is a very important issue.

In this challenge, images of white blood cells had to be sorted into 13 different classes. To train the model, 28.9k labeled images were provided. The goal was then to correctly predict the labels of an additional 9.6k images.

The main challenge here is that the images can be heavily degraded by noise corruption and color shift, while background cells can occupy a large part of the image and hide useful information. An additional issue comes from the class distribution (see Table 1): the class *SNE* represents 45.03% of the whole dataset, while classes like *PC* or *PLY* represent only 0.24% and 0.04%, respectively.

Name	Label	Image Count	Share of dataset
Segmented neutrophil	SNE	13 015	45.03%
Lymphocyte	LY	8 101	28.03%
Monocyte	MO	2 746	9.50%
Blast cell	BL	2 012	6.96%
Eosinophil	EO	861	2.98%
Myelocyte	MY	441	1.53%
Basophil	BA	415	1.44%
Band-form neutrophil	BNE	391	1.35%
Variant (atypical) lymphocyte	VLY	366	1.27%
Metamyelocyte	MMY	360	1.25%
Promyelocyte	PMY	114	0.39%
Plasma cell	PC	68	0.24%
Prolymphocyte	PLY	11	0.04%
Total	-	28 901	100%

Table 1: Image repartition per class

1.2. Main idea of the method

Our solution is based on a two-stage approach. First, as a preprocessing step, we segment the cell of interest to isolate it. This stage removes background objects such as other cells, as well as noise.

The second step is a supervised auto-encoder whose goal is to create a compact latent representation from the segmented image. This representation is trained to satisfy two objectives simultaneously: it must preserve enough information to reconstruct the input image, while also being discriminative for cell-type classification.

1.3. Key contributions

The main contributions of our method are threefold. First, we use GrabCut segmentation as a preprocessing step to isolate the cell of interest and reduce the impact of background objects and noise. Second, we learn an interpretable latent representation that can be analysed with tools such as PCA, which helps us understand how the images are organized in feature space. Finally, we use a shared latent representation for both reconstruction and classification, so image features and class information are learned jointly.

2. Method

2.1. Preprocessing

Before feature extraction, a preprocessing stage is applied to isolate the white blood cell of interest and normalize its appearance. This stage is not limited to producing a binary mask: it is used to detect the relevant cell in a cluttered microscopy image, suppress distracting structures such as red blood cells or staining artifacts, and generate a more stable reconstruction target for the auto-encoder. The pipeline is heuristic rather than learned, which is well suited to this setting since no pixel-level segmentation annotations are available and the dataset exhibits strong priors: the target cell is usually near the image center and its nucleus is often visually distinctive.

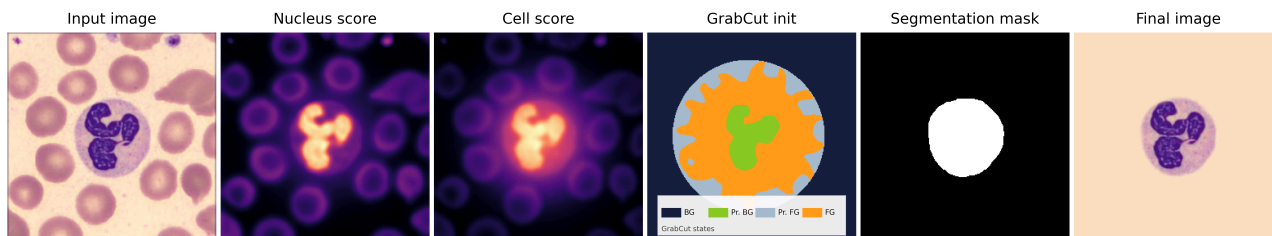


Figure 1: Overview of the preprocessing pipeline

2.1.1. Scoring the nucleus and the cell body

The segmentation starts by detecting the nucleus rather than the full cell, since it provides a more reliable anchor. After Gaussian smoothing, the image is converted to HSV and a nucleus score is computed by combining saturation, darkness, and a spatial prior centered on the image. Candidate regions are extracted by thresholding this score map, cleaned morphologically, and ranked according to their average score, compactness, and distance to the image center. This yields a robust nucleus seed even in the presence of distractors.

Once the nucleus has been identified, a broader cell score is estimated to capture the likely extent of the full white blood cell. This second score combines color cues associated with purple-stained regions, saturation, and the same central prior, while penalizing red responses in order to suppress red blood cells and similar artifacts. The nucleus score therefore provides a reliable localization cue, while the cell score gives a softer estimate of where the cytoplasm may extend around it.

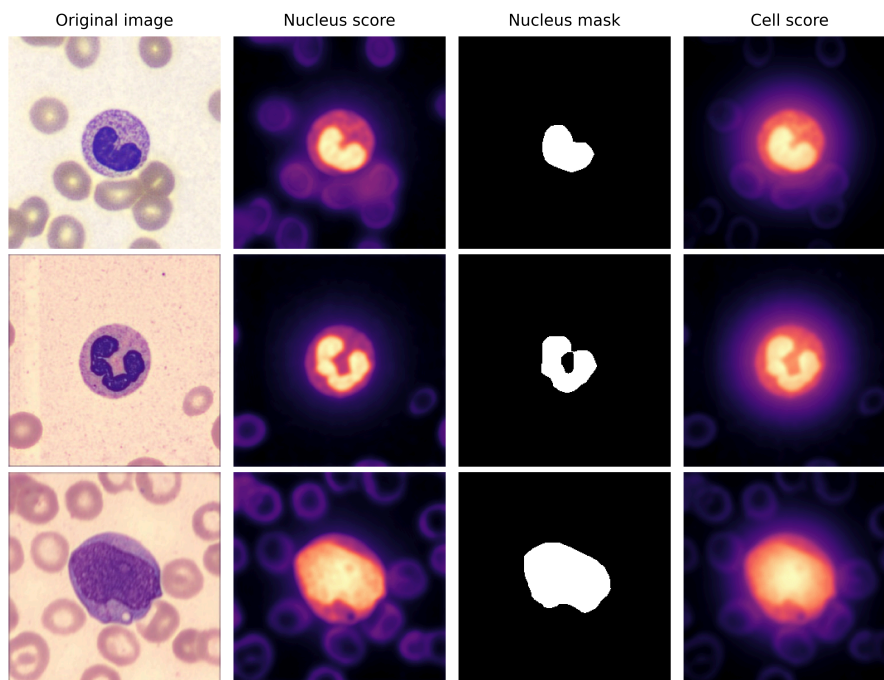


Figure 2: Nucleus and cell scoring used to initialize the segmentation

2.1.2. GrabCut-based segmentation and normalization

The final segmentation is obtained with GrabCut, initialized not from a rectangle but from the previously computed nucleus and cell scores. The detected nucleus is used as sure foreground, while pixels that are unlikely to belong to the target cell are marked as background according to both their distance from the nucleus and

their score values. This custom initialization allows GrabCut to expand the nucleus into a full-cell mask while remaining constrained by both geometric and photometric priors.

The resulting mask is then cleaned by selecting the most plausible connected component and applying small morphological refinements. Finally, the segmented cell is translated so that its centroid lies at the image center, and the removed background is replaced by a color estimated from the pale regions of the original image. A soft blending is applied along the mask boundary to avoid hard cut-and-paste edges. The output of preprocessing is therefore not only a segmentation mask, but also a centered and normalized image that is more stable for the downstream model.

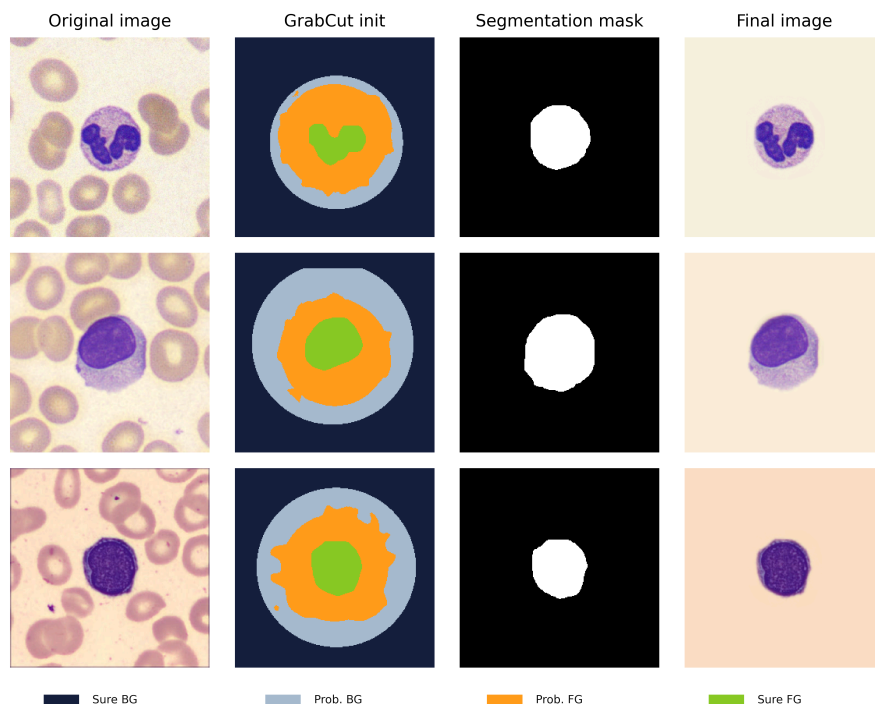


Figure 3: *GrabCut refinement and final normalized segmentation*

2.1.3. Role and limitations

This preprocessing is an important part of the learning pipeline. It reduces irrelevant variation due to distractor cells, background texture, and object position, and encourages the encoder to focus on the morphology of the target white blood cell. In addition, the generated mask is later reused to guide the reconstruction objective, so the segmentation is not only a visual cleanup step but also a structural cue for training.

The main strength of this approach is that it exploits dataset-specific priors without requiring manual segmentation labels. However, it remains a hand-engineered heuristic method: it depends on fixed thresholds and weights, assumes that the cell is reasonably central, and may fail when staining is atypical or when the nucleus is not clearly visible. Despite these limitations, it provides a practical way to simplify and normalize the classification problem.

2.2. Model

The proposed network is not a plain classifier. It is a joint autoencoder-classifier trained on the segmented cell images produced by the preprocessing pipeline. The central idea is to learn a shared latent representation that is useful both for classification and for reconstruction. In practice, the classification branch encourages the latent space to remain discriminative, while the reconstruction branch acts as a structural regularizer and encourages the embedding to preserve meaningful visual information about the cell.

The model takes as input the segmented and normalized cell image rather than the raw microscopy crop. During training, it predicts both a class label and a reconstructed segmented image. The segmentation mask is also used during training to weight the reconstruction loss, so that the network focuses primarily on the cell itself rather than on the uniform background.

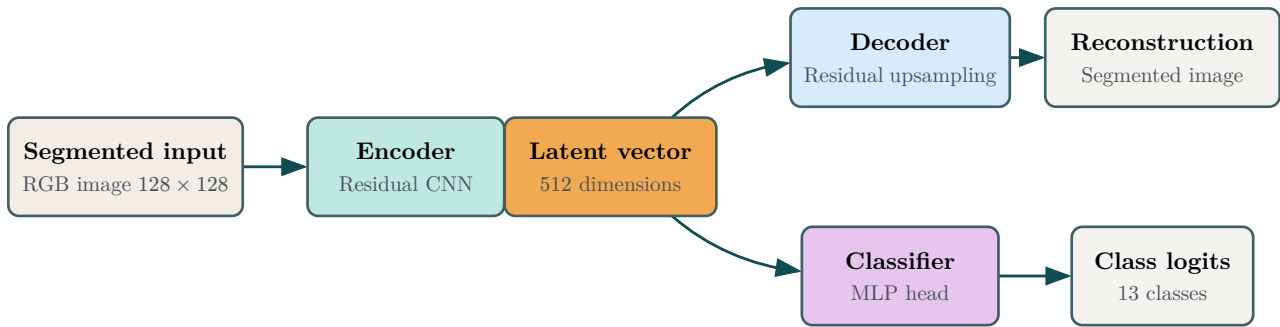


Figure 4: Overview of the joint autoencoder-classifier architecture

2.2.1. Shared latent representation

The model is composed of three parts: an encoder, a decoder, and a classifier head. The encoder maps the input image to a latent vector of dimension 512, which forms the bottleneck of the architecture. This latent code is shared by the two downstream branches: the decoder reconstructs the segmented image, while the classifier predicts the class logits. The architecture is therefore a multitask model with a shared representation.

This design is meant to prevent the encoder from learning features that are only useful for a narrow classification shortcut. By forcing the same embedding to support reconstruction, the latent space is encouraged to retain a more structured description of the cell morphology and appearance. This property is particularly useful in this project, since one of the goals is not only to classify the cells but also to analyze and interpret the learned representation.

2.2.2. Encoder, decoder, and classifier

The encoder is a convolutional residual network operating on 128×128 segmented RGB images. It progressively reduces the spatial resolution through four downsampling stages while increasing the number of feature channels, before applying global average pooling to obtain the final 512-dimensional embedding. This produces a compact global representation of the input cell.

The decoder mirrors this compression process in the opposite direction. Starting from the latent vector, it first reconstructs a small spatial feature map and then progressively upsamples it through residual upsampling blocks until it returns to the original image resolution. The final output is a three-channel reconstructed image in $[0, 1]$. The classifier head is a lightweight multilayer perceptron applied directly to the latent vector. Its role is deliberately simple, so that most of the representational burden remains in the shared convolutional encoder rather than in a large fully connected classifier.

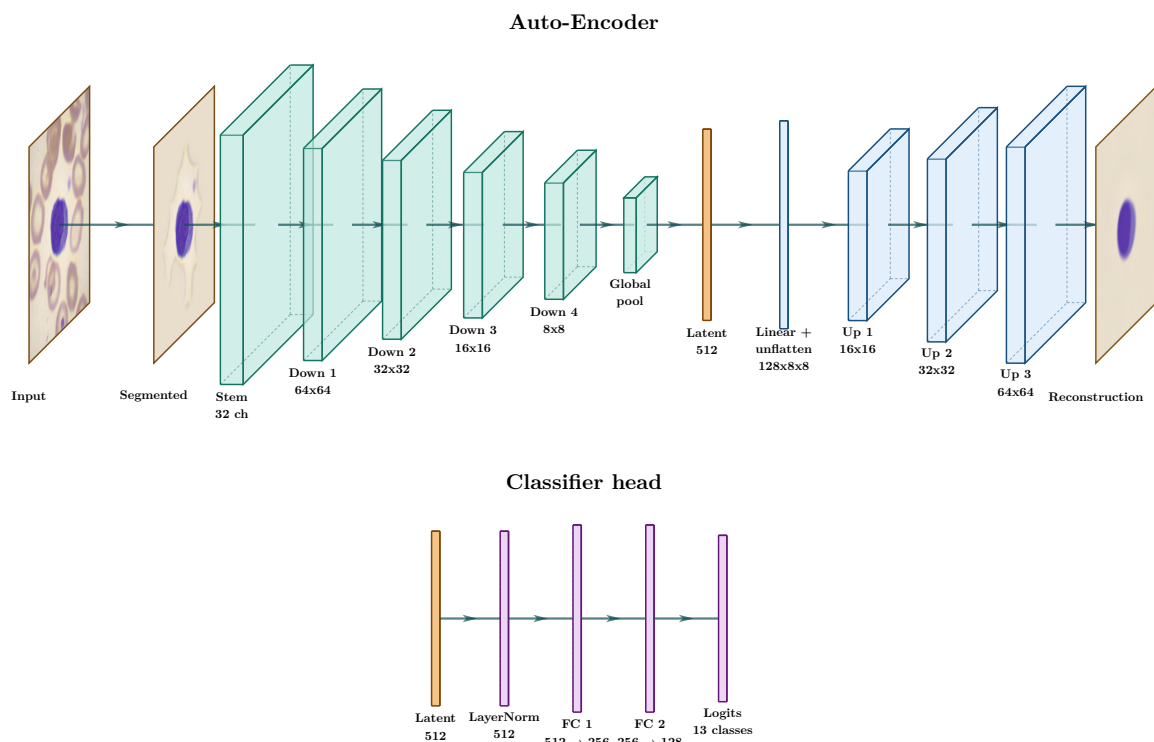


Figure 5: Detailed structure of the encoder, decoder, and classifier head

2.2.3. Joint training objective

The model is trained with two losses: a reconstruction loss and a classification loss. The reconstruction term is a mask-weighted L1 loss between the predicted image and the clean segmented target. Pixels belonging to the segmented cell receive a higher weight than background pixels, which makes the reconstruction objective focus on the biologically relevant foreground. The classification term is a cross-entropy loss applied to the class logits, with both label smoothing and class weighting to better handle class imbalance.

This joint objective makes the model learn a latent space that is simultaneously descriptive and separable. In addition, the training pipeline uses substantial data augmentation, including random crops, flips, rotations, blur, noise, and patch dropout. Patch dropout is particularly relevant here because missing regions are replaced by the estimated background color rather than by black patches, which keeps the corruption visually consistent with the segmented data distribution.

2.2.4. Training strategy and limitations

Optimization is performed with AdamW, and model selection is based on macro F1 on a stratified validation split. This metric is well suited to the challenge because it gives equal importance to all classes despite class imbalance. For the training procedure we tried including class-weighted sampling whose strength increases progressively over time, which provides a conservative way to rebalance the data without overcorrecting from the first epochs. But using weighted cross entropy also performed quite well.

The main strength of this model is that it matches the structure of the overall pipeline: the preprocessing isolates the target cell, and the network then learns a compact representation constrained by both recognition and reconstruction. However, this design also introduces tradeoffs. The decoder increases training complexity even though it is not needed at inference time, and the whole approach depends strongly on the quality of the upstream segmentation. More fundamentally, it assumes that the features useful for reconstructing the segmented cell are also relevant for classification. While this is a reasonable assumption in this setting, it is not guaranteed in general.

3. Results

3.1. Quantitative classification performance

With this setup, we obtained a high classification accuracy on the validation set, which represented 20% of the labeled dataset. During training, the model reached up to 96% validation accuracy, showing that the learned representation generalized well to unseen samples.

Reaching this level of performance, however, required a careful control of model capacity. In particular, strong dropout had to be applied in the classification head (0.35 and 0.45 on the first and second hidden layers, respectively), and the overall architecture had to remain relatively small. Larger models consistently led to significantly stronger overfitting (see the Figure 6).

Despite this strong accuracy, the macro F1 score remained less satisfactory. This reflects the strong class imbalance of the dataset: the model performs well overall, but remains less reliable on the rarest classes. In this respect, a pretrained architecture such as ResNet18 would likely have provided a stronger baseline. Unfortunately, the available computational resources were not sufficient to explore this direction properly.

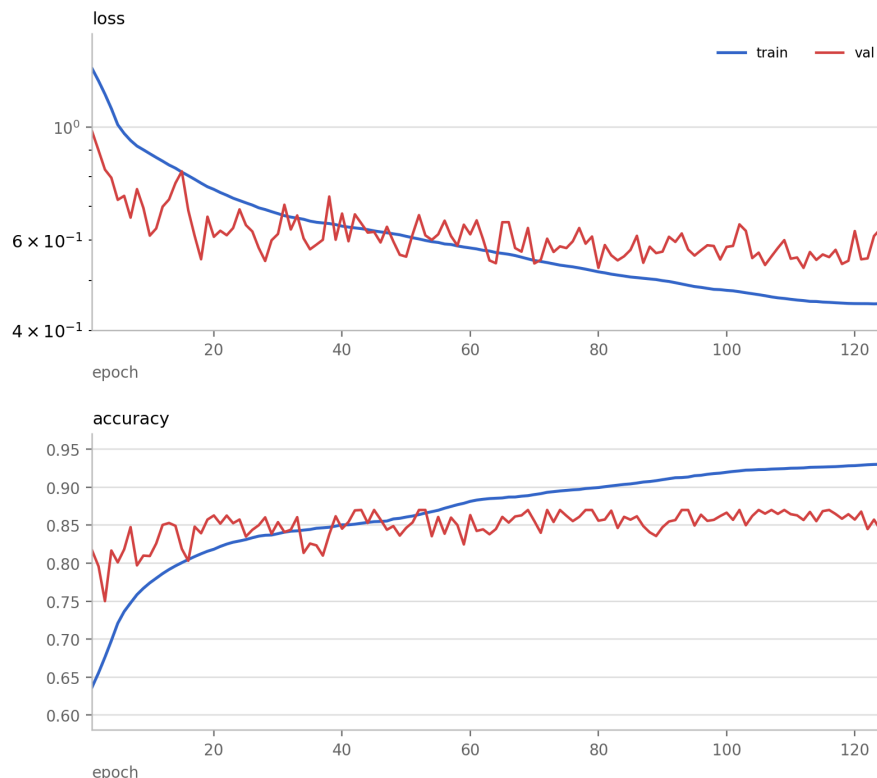


Figure 6: Overfitting on the training set (the loss and accuracy stop improving early) using a larger classifier

3.2. Impact of preprocessing and multitask learning

We also conducted several ablation experiments to assess the contribution of the main components of the pipeline. The first test consisted in removing the preprocessing stage entirely and feeding the raw images directly to the model. In that setting, the model struggled to reconstruct the input, as the numerous background red blood cells and artifacts introduced a large amount of irrelevant visual complexity. A significant part of the latent space was therefore wasted modeling nuisance structures rather than the target cell itself.

We then evaluated an intermediate setting in which the raw image was used as input, while the segmented image was kept as the reconstruction target. This produced better results than the fully raw setting, but the model still spent many unnecessary epochs implicitly learning to isolate the cell before solving the actual classification problem. In the end, the performance remained comparable to directly using the segmented image as input, while requiring a more difficult optimization process.

We also compared different background replacement strategies. Before adopting the current light background estimated from the image itself, we first used a black background outside the segmented cell. In practice, this introduced overly sharp transitions at the cell boundary and created unrealistic visual artifacts. Preserving background color information therefore proved important for stabilizing both the reconstruction task and the learned representation.

To validate the multitask design, we also removed the reconstruction branch entirely. In that case, the model became closer to a small residual classifier, still much lighter than standard architectures such as ResNet. This classifier-only version reached about 84% validation accuracy, but the learned features were no longer interpretable since they could no longer be decoded back into image space. More importantly, this experiment showed that the reconstruction branch provides a strong regularizing effect. We also tested sequential training, by first training the autoencoder and then replacing the decoder with the classification head, but this approach caused the latent representation to lose most of its visual meaning.

3.3. Reconstruction quality and latent space analysis

As shown in Figure 8, the model learns a latent representation that is rich enough to reconstruct the segmented input image. Even in difficult cases, such as the last example in the figure, the reconstructed output remains visually close to the target. More generally, although the reconstructions are smoother than the original inputs, the main morphological attributes are well preserved, in particular the global cell shape, the nucleus shape, and the main color distribution.

This reconstruction capability makes it possible to analyze the latent features visually. For instance we compared the average image with the reconstruction of the average latent encoding, see Figure 7.

For another analysis, we encoded the full dataset and applied PCA to the resulting latent vectors. This provides a compact visual representation of the dominant modes of variation learned by the model. Compared with performing PCA directly on image pixels, this approach has two major advantages: the dimensionality is already much lower, and the latent features are explicitly trained to remain discriminative through the classification branch.

Using the six principal directions with the highest explained variance (see Figure 9), we observe that the first component, which explains 41.9% of the variance, appears to capture several important morphological properties of the nucleus: its shape, its relative size with respect to the full cell, and whether its central region appears hollow or dense. The second component is mainly related to the overall size of the cell, similarly to the fourth component. The remaining components are more strongly associated with color variations in the cell interior and in the estimated background.

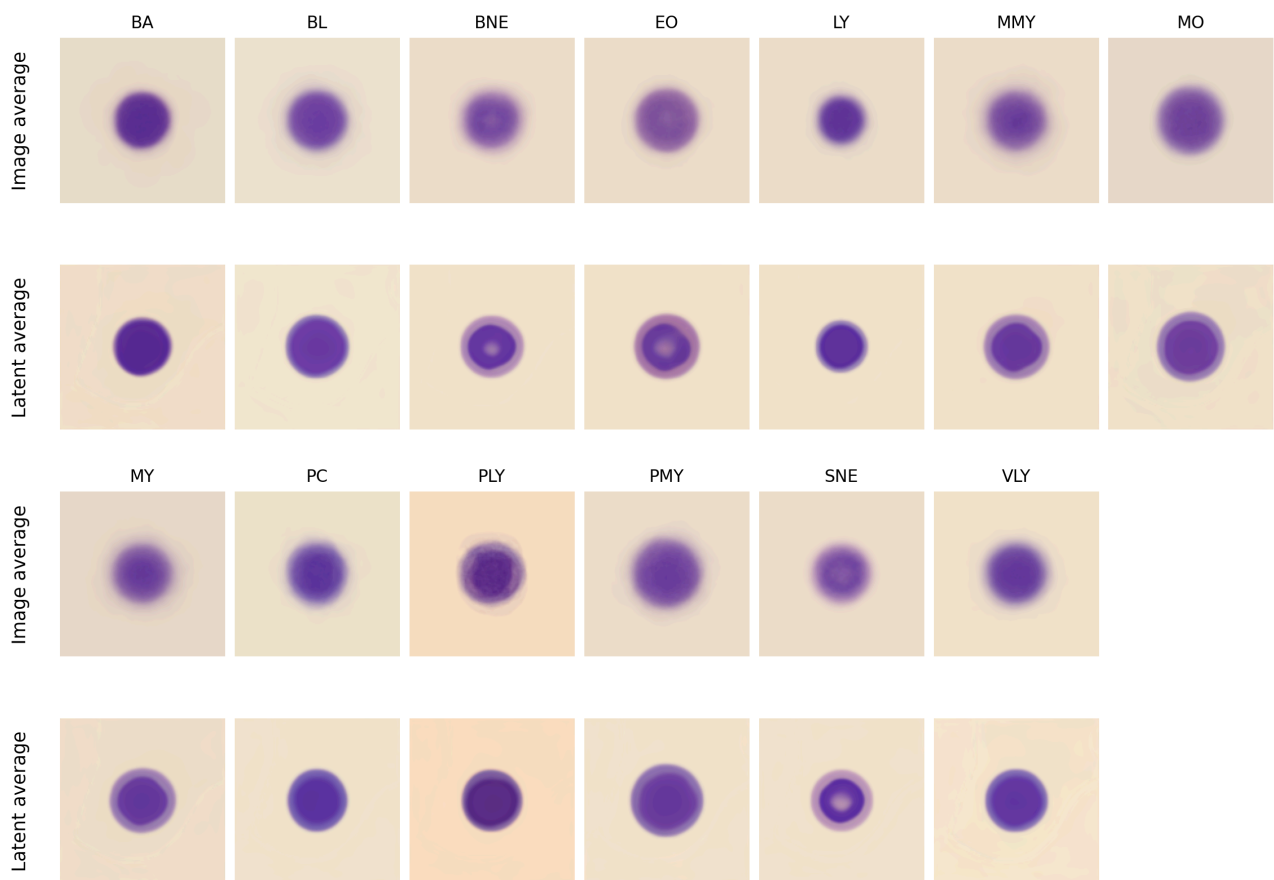


Figure 7: Comparison between the average segmented image and the decoded average latent representation

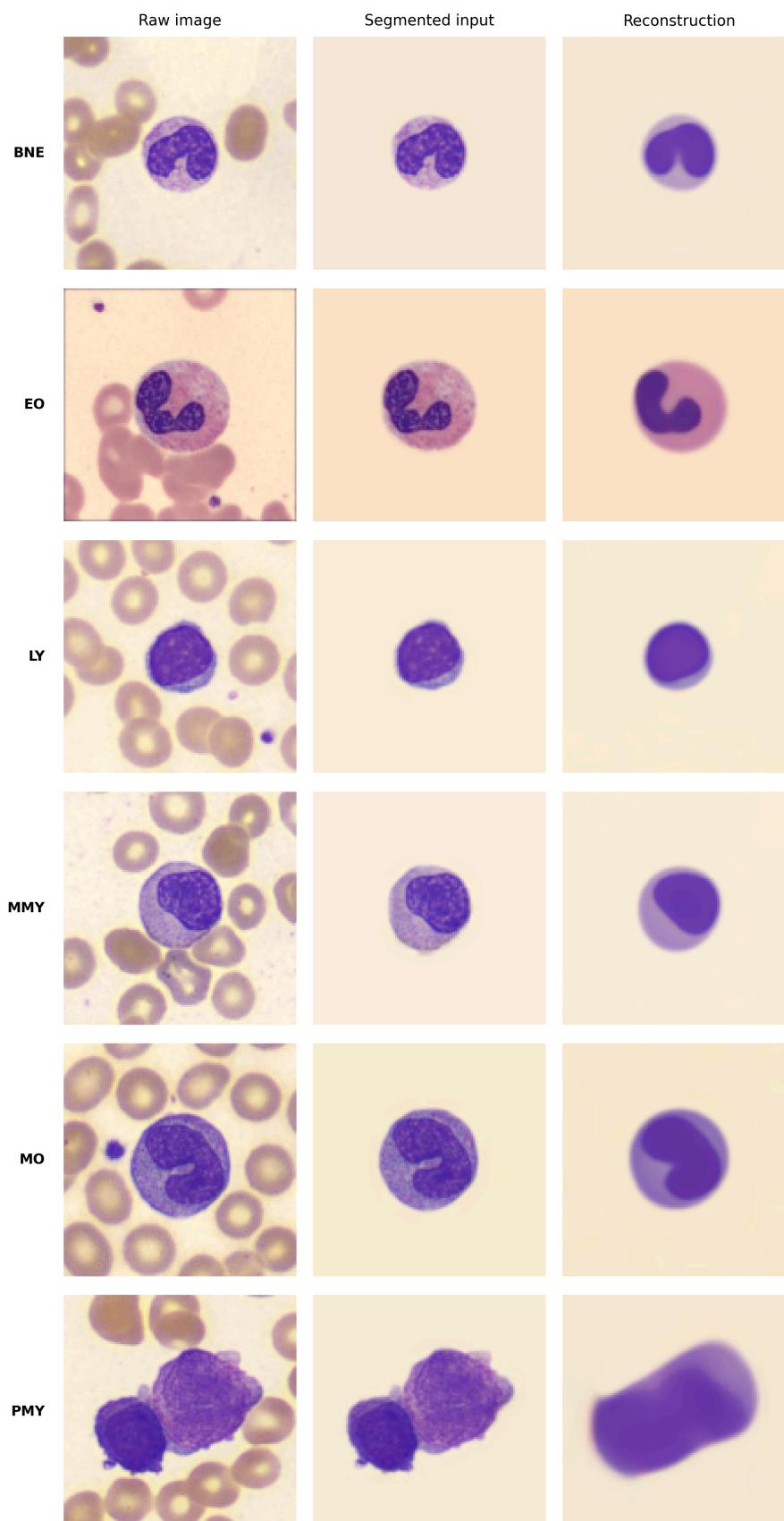


Figure 8: Reconstruction of the segmented images

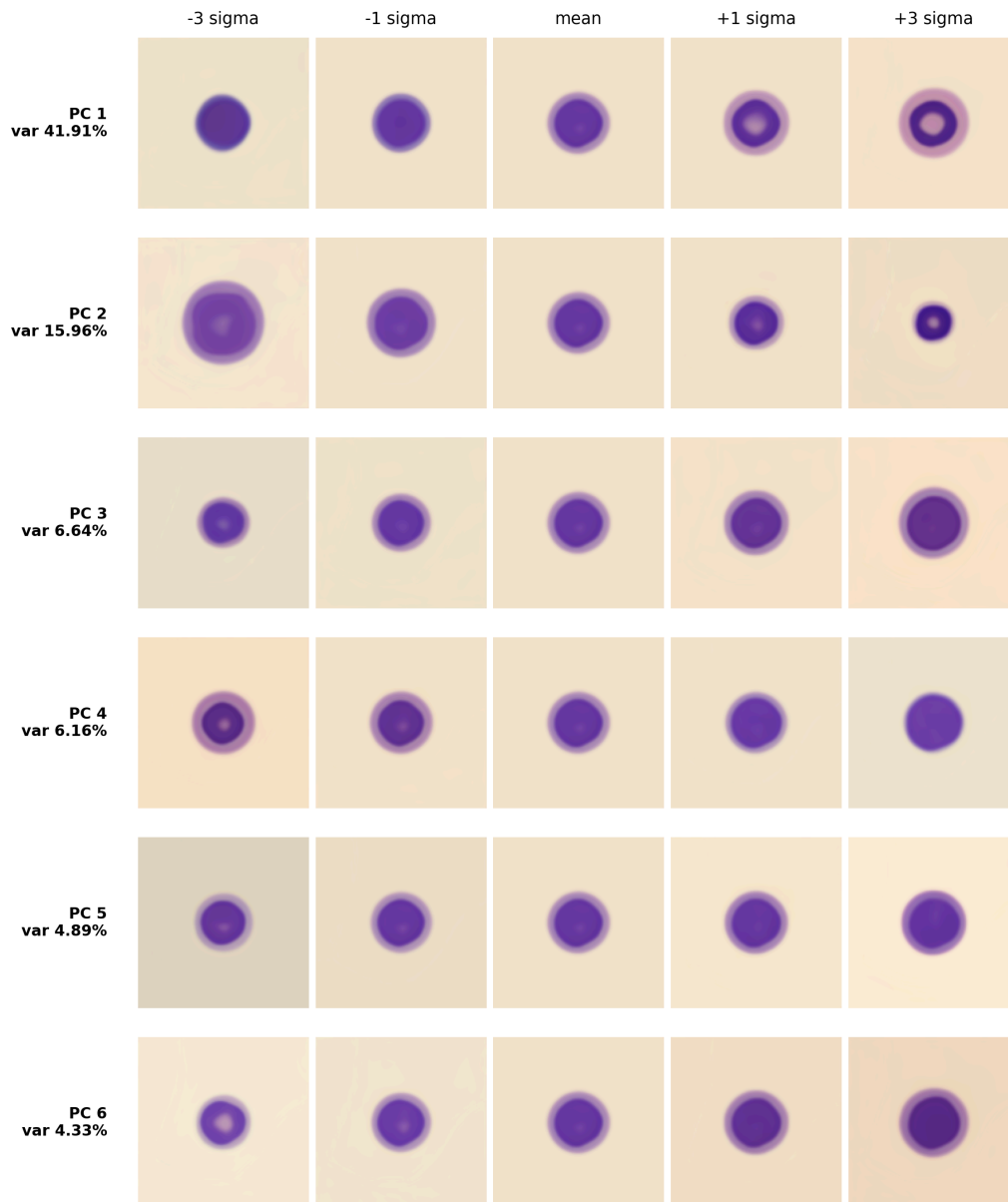


Figure 9: Variations along the six principal latent directions

3.4. Error analysis and limitations

Although the proposed model is a useful analysis tool and performs well in terms of global accuracy, it remains limited by the strong class imbalance of the dataset. In particular, it has more difficulty predicting the rarest classes reliably, which explains the gap between accuracy and macro F1.

In addition, the method depends strongly on the quality of the preprocessing stage. When the segmentation is imperfect, the model may receive an incomplete or misleading representation of the target cell. More generally, the joint reconstruction-classification objective improves structure in the latent space, but it also makes the training pipeline more constrained than a standard classifier.

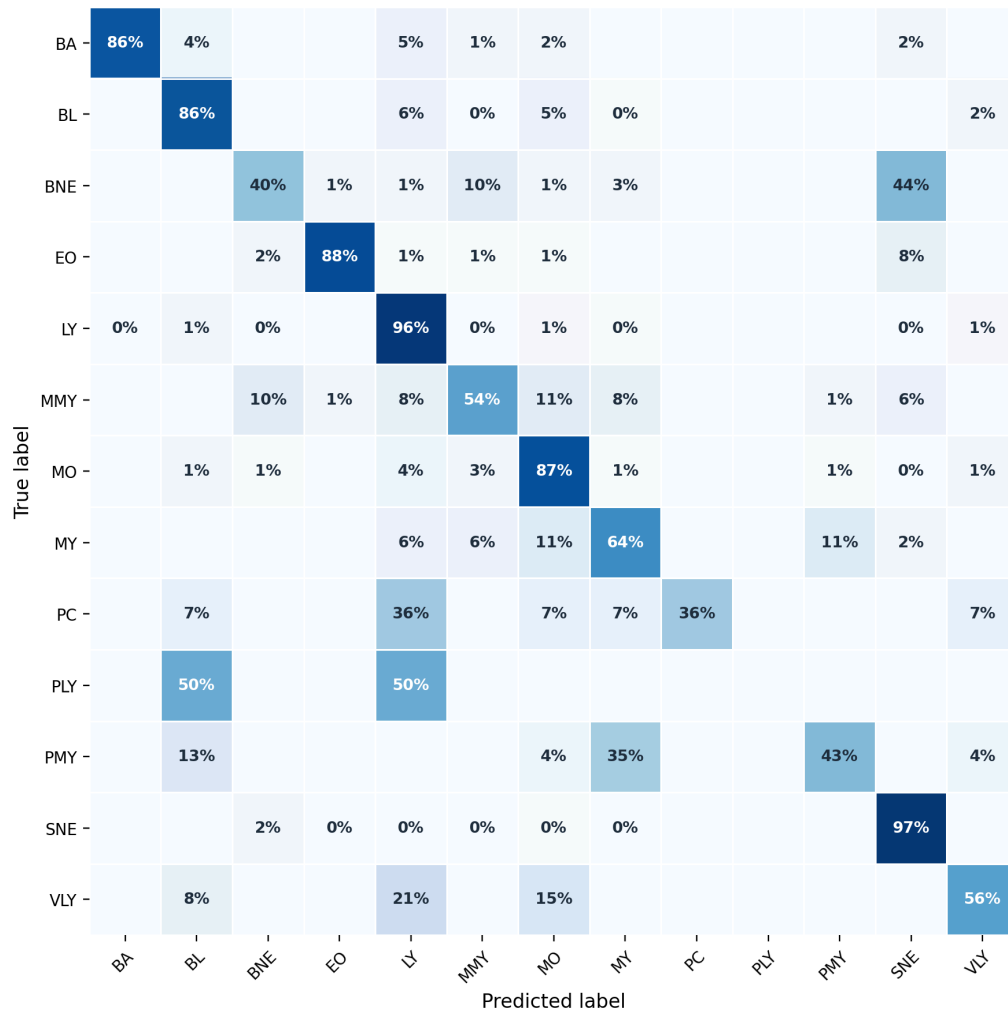


Figure 10: Confusion matrix on the Validation Set

4. Conclusion

In this work, we proposed a cell classification pipeline combining heuristic preprocessing with a joint autoencoder-classifier architecture. The preprocessing stage isolates, recenters, and normalizes the target white blood cell, which reduces irrelevant variation in the input and simplifies the downstream learning problem. The model then learns a shared latent representation constrained by both classification and reconstruction, encouraging the features to remain discriminative while preserving meaningful visual structure.

The experiments show that this approach achieves strong classification performance and that both preprocessing and multitask learning contribute significantly to the final results. Beyond classification alone, the reconstruction branch also makes it possible to analyze the latent space and relate its dominant directions to visible morphological variations in the cells. Although the method remains limited by class imbalance, heuristic segmentation, and the available computational resources, it provides a coherent and effective framework for learning interpretable representations for cell classification.